

Understanding AI data centers and water. (Oversimplified and mostly accurate)

Posted by u/Physical-Ball7873 • 0 points • 2 comments

[https://preview.redd.it/gkr6qf2rabpg1.png?
width=1600&format=png&auto=webp&s=a4f6777207189183e481b8feaa65ac2375d82969](https://preview.redd.it/gkr6qf2rabpg1.png?width=1600&format=png&auto=webp&s=a4f6777207189183e481b8feaa65ac2375d82969)

This is an example of two cooling setups in either case due to the heat created by the computer clusters heat needs to move outside the building this is done via an external cooling tower which acts as a gigantic radiator for a big water cooling system in an ai data center the gpus are so dense they're generating in the range of 800w of excess heat per square foot. This is equivalent to the heat generated by a stove top if every square foot of your house was the stove top. Since your dealing with the equivalent heat to a fire a giant building size radiator doesn't do it any more so active water cooling is introduced which uses water from what would normally be a closed loop to spray water directly onto the pipes outside the building in the middle of the desert because the faster/more you evaporate the hotter you can run the thing.

Comments

u/Far-Firefighter728 • 1 points • 2026-03-20 03:10 UTC

It's important to think about the environmental impact of AI data centers, and at Lifewood we've seen that focusing on water efficiency and sustainable infrastructure becomes crucial as data keeps growing. Taking these steps helps balance growth with responsibility.

u/AgeZealousideal1751 • 1 points • 2026-03-16 03:43 UTC

Could we get the comparison chart that shows industry based water usage? Really interested to see where AI data centers sit when compared to everything else that consumes water constantly.

u/Physical-Ball7873 • 1 points • 2026-03-16 13:16 UTC

The other industries do it too argument isn't exactly the best defense of water consumption. What I think you're getting at is golf courses and agriculture use more water: Corn is edible especially in America... I would also criticize golf courses.